

Evolution: High-Yield Important Diagrams & Schematic Charts

Board Exam Oriented Vector Schemes • Anatomical Diagrams •
Experimental Setups • Labelling Keys

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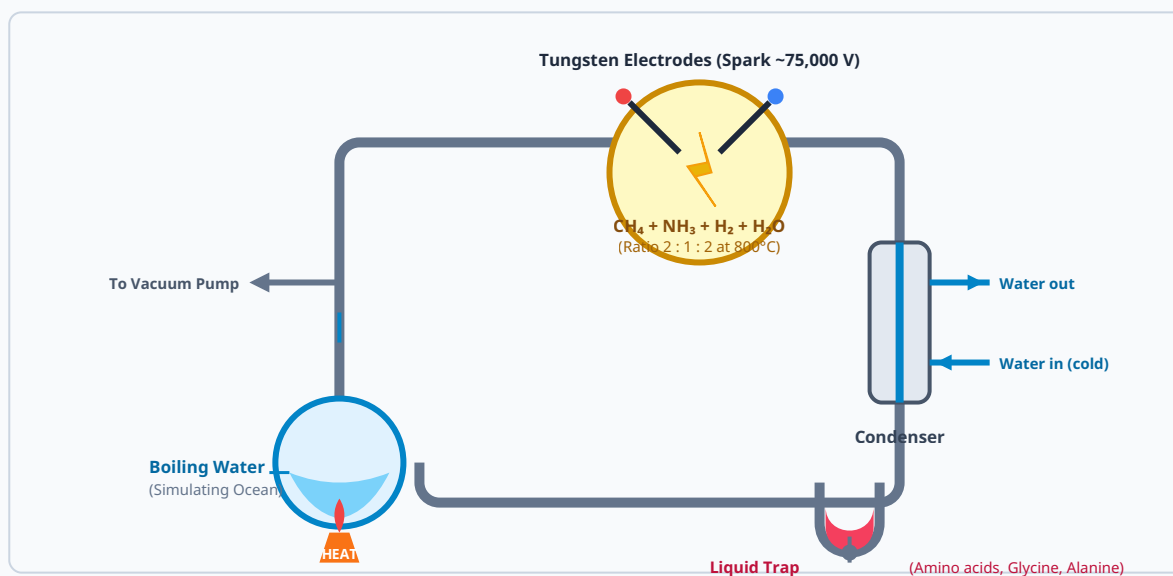
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DIAGRAM 1: STANLEY MILLER'S SPARK DISCHARGE APPARATUS (5 MARKS BOARD ESSENTIAL)

Experimental setup simulating prebiotic Earth conditions (1953). Recreated a reducing prebiotic atmosphere with CH_4 , NH_3 , H_2 (2:1:2) and boiling water steam subjected to high voltage electric sparks at 800°C .



Exam Scoring Checklist: Check that you have marked: (1) Spark discharge chamber at 800°C , (2) Boiling flask simulating the ocean, (3) Cold water condenser, (4) Ratio of gases $\text{CH}_4 : \text{NH}_3 : \text{H}_2 = 2:1:2$, and (5) Trap containing synthesized racemic amino acids.

DIAGRAM 2: LOUIS PASTEUR'S SWAN-NECK FLASK EXPERIMENT (DISPROOF OF ABIOTENESIS)

Stepwise experimental demonstration of **Biogenesis** (1864). Curved neck allowed air exchange while gravitationally trapping dust and airborne bacteria in the S-bend.

DIAGRAM 3: THREE OPERATIONAL MODES OF NATURAL SELECTION (3 MARKS BOARD ESSENTIAL)

Phenotypic response curves showing shifts under different environmental selection pressures: **Stabilizing** (intermediate favored), **Directional** (one extreme favored), and **Disruptive** (both extremes favored).

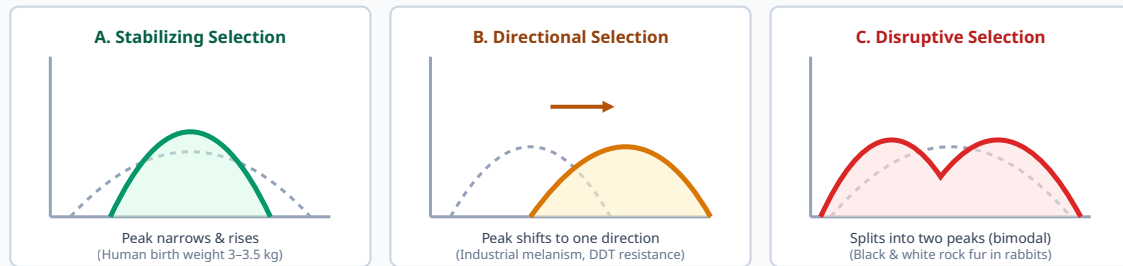


DIAGRAM 4: ANATOMICAL EVIDENCE — HOMOLOGY VS. ANALOGY COMPARISON

Structural comparison showing pentadactyl limb conservation in mammals (Homology → Divergent Evolution) versus convergent optical design in Octopus and Mammalian eyes (Analogy → Convergent Evolution).



DIAGRAM 5: ARCHAEOPTERYX LITHOGRAPHICA ANATOMICAL SCHEME (CONNECTING LINK)

Fossil discovered in Solnhofen limestone, Jurassic period. Clearly illustrates evolutionary transition bridging Reptilia and Aves.

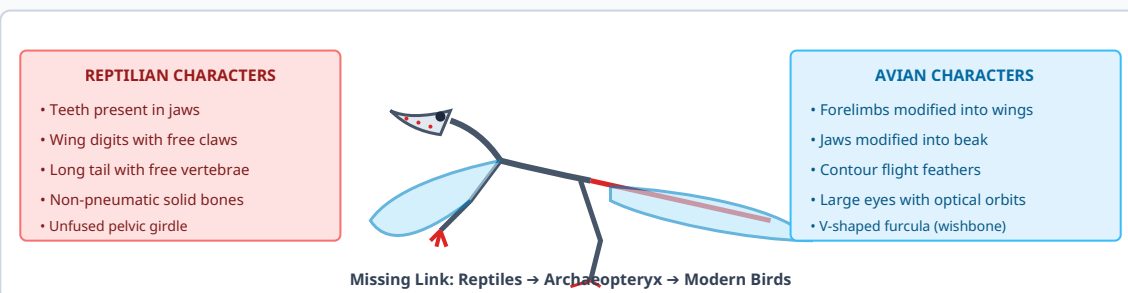
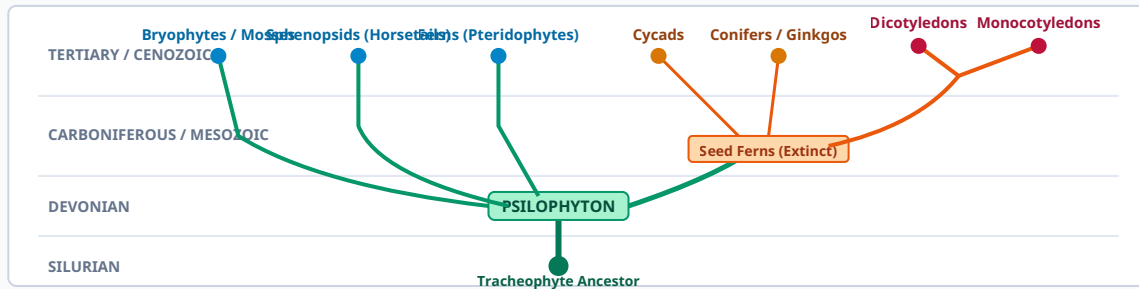


DIAGRAM 6: PLANT EVOLUTIONARY LINEAGE THROUGH GEOLOGICAL ERAS (PSILOPHYTON TREE)

Phylogenetic tree tracing terrestrial vascular plant diversification from primitive Silurian tracheophytes and *Psilophyton* into modern Angiosperms and Conifers.



Key Landmark: *Psilophyton* (originating in Devonian) is the stem ancestor linking Sphenopsids, Ferns, and Seed Ferns (which gave rise to Gymnosperms and Angiosperms).

DIAGRAM 7: HOMINID CRANIAL CAPACITY EVOLUTION & POSTURE LINEAGE

Progressive cranial enlargement (cc), facial flattening (orthognathism), and bipedal refinement across hominid ancestors.

